MATTHEW EDGAR

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EDUCATION

University of California, Berkeley **May 2026**
Bachelor of Science in Chemical Biology – Concentration in Computational Chemistry

San Diego Mesa College **May 2024**
Associate of Science in Chemistry
Associate of Arts in Mathematics and Pre-Engineering Mathematics
Graduated with Honors

Relevant Coursework:
- **Programming:** C/C++ Programming, Python Programming for Engineers, Computational Chemistry, Numerical Analysis
- **Mathematics:** Discrete Mathematics, Mathematical Tools for the Physical Sciences, Linear Algebra, Differential Equations

EXPERIENCE

Williams Research Lab – University of California, Berkeley **January 2025 – May 2026**
Undergraduate Research Assistant
Berkeley, CA
- Participated in the design and assembly of a nanoelectrospray apparatus, including custom circuitry and resistive components.
- Conducted 100 nanoelectrospray ionization experiments and developed Python scripts to collect and analyze their data.

LIFEwest Ambulance Company **October 2024 – June 2025**
Emergency Medical Technician
Oakland, CA
- Transported injured and critically ill patients to and from medical facilities across the San Francisco Bay Area.
- Logged over 500 hours in time-sensitive clinical environments.

T-Mobile **October 2021 – May 2022**
Mobile Expert
San Diego, CA
- Assisted customers with mobile devices, wireless plans, and troubleshooting while providing individualized connection plans.
- Processed sales, device upgrades, activations, and account changes while meeting customer-service and sales objectives.

United States Navy **June 2017 – June 2021**
Machinist's Mate (E-5)
San Diego, CA
- Active-duty enlisted sailor in the engineering departments of two ships: the USS Ardent and the USS Makin Island.
- Operated and maintained shipboard engineering systems, including refrigeration, diesel engines, and reverse-osmosis systems.
- Maintained fuel and oil systems, weapons systems, and kept proper watchstanding logs for quality assurance.

PROJECTS

Bitcoin Seed Phrase Unscrambler
Python
- Takes a partial BIP39 seed phrase between 12-24 words and identifies which word positions are known, missing, or out of order via command line arguments.
- Calculates the estimated compute needed to complete the seed phrase and determines if it is statistically feasible.
- Attempts every possible combination of the seed phrase, verifies its checksum, and accesses it on the blockchain.

Monte Carlo Sampler Using Egg Box Potential
Python
- Created a Python script that uses the Monte Carlo method to estimate the egg box energy integral of a particle.
- This project further applies different forces to the particle and simulates the positions of two interacting particles.

Personal Website
HTML, CSS, GitHub Pages
- Developing a personal portfolio website to display my coursework and software projects.

TECHNICAL SKILLS

Programming Languages: Python, C++, MATLAB
Experienced in: NumPy, Matplotlib, Monte Carlo Methods, Numerical Estimation Methods, Microsoft Office Suite
Tools: Git, GitHub, VS Code, Jupyter Notebook
Languages: English (native), Spanish (conversational)